SDG 13: Climate Action

Inclusive Climate Risk Communication through Language Access

Introduction

Sustainable Development Goal 13 (SDG 13): Climate Action aims to take urgent action to combat climate change and its impacts. It focuses on strengthening resilience and adaptive capacity to climate-related hazards, integrating climate action into planning, improving education and awareness, and supporting vulnerable communities in climate adaptation.

This use case focuses on climate risk communication, early warning comprehension, preparedness, adaptation literacy, and community-level climate action in Sri Lanka. It is intentionally framed around climate-related information access rather than poverty relief, energy access, water safety, or city-level municipal communication.

In Sri Lanka, communities increasingly experience floods, irregular rainfall, heat, drought-related stress, landslide risk, and other climate-related hazards. Recent research from Gampaha shows that women and girls clearly recognize environmental changes such as increased flooding, irregular rainfall, and heat, even when many are unfamiliar with the term climate change. This shows a critical gap between lived climate experience and formal climate literacy.

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This report explores how VosynCore can support SDG 13 by strengthening multilingual climate communication. Through voice-based guidance in Sinhala, Tamil, and English, VosynCore can help households, women, youth, farmers, health workers, local officers, and community groups understand climate warnings, preparedness steps, adaptation options, emergency instructions, and climate education materials.

Modelled pilot projections suggest that this climate-risk communication architecture could ***improve warning comprehension by up to 35-45%, increase preparedness-task completion by up to 30%, improve user confidence in climate adaptation guidance by up to 25-35%, and reduce communication-related delays during climate-related emergencies by 15-25% among participating communities.***

Use Context: Climate Risk, Preparedness & Adaptation Communication

Climate action in Sri Lanka depends not only on national policies and technical climate data, but also on whether communities receive practical, trusted, and locally understandable guidance before, during, and after climate-related hazards. Early warnings, climate education, and adaptation messages can only protect people when they are understood and acted upon.

A central challenge for SDG 13 is the gap between climate risk information and community action. People may hear about heavy rainfall, flood risk, landslide warnings, heat, drought, or adaptation programmes, but still remain uncertain about what those warnings mean, what actions should be taken, where to seek help, or how to prepare their household.

The Gampaha flood adaptation study shows this gap clearly. Many women and girls had limited understanding of the term climate change, yet they could describe practical environmental changes such as heavier rainfall, more frequent flooding, and increased heat. They also reported disruption to income, education, health, sanitation, caregiving, and household safety during floods.

> **Example:** A family may hear that heavy rainfall is expected, but may not know whether to move documents to higher shelves, prepare dry food, check medication, identify a safe route, protect school materials, or help older relatives relocate before floodwaters rise.

Community-based adaptation research also shows that Sri Lanka needs stronger grassroots participation in climate planning. National adaptation policies exist, but local engagement can be limited by top-down governance, low awareness, weak decentralization, and a lack of formal participation structures. This creates a gap between climate planning and the everyday realities of vulnerable communities.

Traditional responses such as printed notices, technical reports, one-way awareness campaigns, and English-heavy climate materials are useful but not always enough. Communities need clear, interactive, and trusted communication that explains climate risks in everyday language and allows people to ask follow-up questions in the language they use at home.

Use Case Narrative: Amara's Journey from Climate Risk to Preparedness

To illustrate the human stakes and potential solutions, consider the story of Amara, a 39-year-old mother living in a flood-prone area of Gampaha with her elderly mother, school-aged daughter, and husband who works irregular shifts.

Amara has lived through several periods of heavy rain and flooding. She knows that rainfall patterns have changed, that some drains overflow faster than before, and that floods disrupt cooking, school attendance, work travel, hygiene, and access to medicine. However, she does not think of these experiences as part of climate change because most climate information she sees is technical or difficult to connect to daily life.

In theory, warning systems and preparedness messages are designed to help residents like Amara. There are weather updates, flood alerts, school closure notices, health messages, and local announcements. But in practice, these messages may arrive through different channels, use formal wording, or fail to explain what action should be taken immediately.

One evening, Amara receives several messages about heavy rainfall and possible flooding. A television announcement mentions low-lying areas. A neighbour shares a social media post. Her daughter receives a school message about possible closure. A local volunteer says the area should prepare, but no one explains the steps clearly.

Amara is worried because her elderly mother needs medication, her daughter has exams, and their important documents are stored in a low cupboard. She wants to prepare, but she is unsure which warnings are reliable, whether evacuation is necessary, and what support will be available if flooding becomes severe.

At the same time, the local health centre and response volunteers are also trying to coordinate. Disaster communication research in Sri Lanka shows that fragmented communication, unclear roles, delayed information flow, outdated contact systems, and limited accountability can weaken emergency response. This means that households and institutions can both struggle when information is not timely, standardized, and understandable.

A turning point comes when a community adaptation pilot introduces VosynCore as a climate-risk communication tool. Amara can speak in Sinhala and ask, 'What should I do before the flood reaches my house?' VosynCore summarizes the warning in plain language, explains immediate household actions, lists documents to protect, reminds her to check medication, and gives guidance on safe movement.

VosynCore also helps the local volunteer translate and simplify preparedness guidance into Sinhala and Tamil voice messages for nearby households. Instead of receiving scattered messages, residents hear consistent instructions on rainfall risk, flood preparedness, safe routes, hygiene precautions, and where to ask for help.

Later, Amara uses the same tool to explain the warning to her mother and daughter. The message is short, spoken, and practical. Her family moves key items to higher shelves, prepares dry food, checks medicine, charges phones, and keeps a small emergency bag ready.

This narrative underscores why VosynCore is relevant to SDG 13. Amara's story is fictional, but it reflects real barriers in climate action: lived experience without climate literacy, warnings without clear steps, and preparedness guidance that is not always delivered in accessible local language.

By replacing fragmented climate messages with localized and interactive voice guidance, VosynCore helps bridge the final-mile communication gap between climate risk information and community preparedness.

Solution Overview: VosynCore's AI-Powered Climate Communication Layer

VosynCore is an AI-driven platform designed to bridge language gaps in real time, enabling users to understand climate risk and preparedness guidance across Sinhala, Tamil, and English. For SDG 13, the platform functions as a multilingual climate communication layer that can support early warnings, adaptation education, community preparedness, and emergency coordination.

This core capability directly addresses the barriers that prevent households and local actors from acting on available climate and disaster information. It supports plain-language explanations, two-

way questions, notice-to-voice conversion, and localized climate education that can be used before, during, and after climate-related hazards.

Key Innovations of VosynCore:

- **Real-Time Climate Warning Translation:** VosynCore can convert rainfall alerts, flood warnings, landslide advisories, heat messages, drought guidance, and evacuation notices into accessible Sinhala, Tamil, and English voice guidance.
    - **Example:** When an early warning message is issued, VosynCore can explain what the warning means, who is most at risk, what actions should be taken now, and where users can seek verified help.
- **Two-Way Conversational Preparedness Support:** Climate action is not a one-way information process. Users often need to ask questions based on household members, location, transport, health needs, livestock, documents, or livelihood activities.
    - **Example:** If a user asks whether to move a sick parent, protect school materials, prepare drinking water, or relocate documents, VosynCore can provide context-aware guidance and recommend verified escalation to local authorities where needed.
- **Notice-to-Voice Climate Literacy:** Many climate materials appear as reports, school resources, posters, PDFs, policy notices, or technical advisories. These formats can be difficult for users with limited time, digital literacy, or technical vocabulary.
    - **Example:** If a community receives a poster about flood preparedness or climate adaptation, the platform can summarize the key actions into a short spoken message in Sinhala, Tamil, or English.
- **Community Adaptation Communication:** SDG 13 requires communities to participate in adaptation planning, not only receive warnings. VosynCore can help translate consultation materials, meeting summaries, household checklists, and local adaptation messages.
    - **Example:** A community organization can use VosynCore to explain why women, youth, older residents, farmers, and low-income households should be included in local climate preparedness planning.
- **Data privacy and Safety:** Climate and disaster guidance can involve location, household vulnerability, health needs, and emergency status. VosynCore must minimize data collection, protect user information, avoid unsupported technical advice, and route urgent cases to verified human responders.

Together, these features make VosynCore a practical climate-action communication tool. It does not replace meteorological agencies, disaster managers, health institutions, or community leaders; instead, it helps users understand and act on verified climate and preparedness information.

Alignment with SDG Targets & Global Frameworks

Through its context-aware AI translation platform, VosynCore addresses communication barriers that prevent communities from understanding climate risks, participating in adaptation planning, and taking preparedness actions before, during, and after climate-related hazards.

SDG 13 Targets Addressed:

Target 13.1: Strengthen resilience and adaptive capacity to climate-related hazards and natural disasters

Resilience depends on whether warnings and preparedness instructions are understood in time. Communities need clear guidance on floods, landslides, heat, drought, evacuation, household preparation, hygiene, and post-event recovery.

- **VosynCore's Role:** By translating and simplifying early warnings and preparedness steps, VosynCore can help households and local responders turn climate-risk information into timely action.

Target 13.2: Integrate climate change measures into policies, strategies, and planning

Climate adaptation planning requires community participation, local knowledge, and clear communication between institutions and residents. Research on community-based adaptation in Sri Lanka highlights the need for stronger grassroots engagement and more inclusive planning structures.

- **VosynCore's Role:** The platform can make climate plans, consultation materials, training resources, and community feedback processes more accessible in Sinhala, Tamil, and English, helping local voices participate in adaptation planning.

Target 13.3 and Target 13.b: Improve climate education, awareness, capacity, and planning for vulnerable communities

Climate literacy and preparedness awareness remain uneven. Evidence from Sri Lankan universities suggests that climate change literacy differs across disciplines, while research with women and girls in Gampaha shows that lived climate impacts are often understood more clearly than the formal language of climate change.

- **VosynCore's Role:** By converting climate education into localized, interactive voice guidance, VosynCore can support awareness, preparedness capacity, and climate learning among students, households, women, youth, older residents, and other vulnerable groups.

Cross-References to Related SDGs

This SDG 13 use case also connects to SDG 1: No Poverty because climate-related hazards can increase economic vulnerability for low-income households. However, SDG 13 focuses on climate risk communication and preparedness, while SDG 1 focuses on disaster relief access and poverty resilience after shocks.

This use case also links to SDG 7: Affordable and Clean Energy because clean energy adoption and energy-saving behaviour can support climate mitigation. However, within SDG 13, the focus

remains on climate awareness, early warning communication, and adaptation preparedness rather than energy access itself.

KPI Mapping & Other Frameworks

We have identified specific KPIs to measure VosynCore's performance and impact. The table below shows how each indicator can connect climate-risk communication, SDG 13 targets, and recognized impact measurement frameworks.

Key Performance Indicator (KPI)	SDG 13 Target & Indicator	Aligned IRIS+ Metric	Other Framework Alignment
Users Reached with Climate Risk Guidance: Number of households, students, women-led groups, community volunteers, and local actors receiving climate warning, preparedness, or adaptation guidance through VosynCore.	**SDG 13.3** Climate education and awareness. **SDG 13.1** Resilience and adaptive capacity to climate-related hazards.	**Client Individuals:** Total number of individuals served through climate-risk communication and preparedness guidance.	**GRI 413-1 (Local community engagement):** Demonstrates community-level climate awareness and preparedness outreach; Sendai Framework indicators on disaster risk awareness and response capacity.
Warning Comprehension Improvement: Percentage increase in users who correctly understand flood, landslide, heat, drought, or rainfall warning messages after using VosynCore.	**SDG 13.1** Strengthen resilience and adaptive capacity. **SDG 13.3** Improve education, awareness, and capacity.	***Outcome Indicator (custom):*** Measures knowledge gain and comprehension quality after intervention.	**GRI 417 (Information and labeling):** Ensures critical information is delivered in understandable formats; humanitarian communication standards on clear risk communication.
Preparedness Task Completion Rate: Percentage of users who complete recommended preparedness actions such as protecting documents, preparing emergency supplies,	**SDG 13.1** Resilience to climate-related hazards and natural disasters.	***Client Outcomes (custom):*** Tracks action taken after guidance, not only message exposure.	**Sendai Framework and Sphere-aligned preparedness principles:** Supports early action, local readiness, and reduced vulnerability.

checking medicines, identifying safe routes, or sharing warnings with family.			
Community Adaptation Participation: Number of users participating in climate adaptation meetings, community feedback sessions, school climate activities, or women-led preparedness discussions supported by VosynCore.	**SDG 13.2** Integration of climate change into planning. **SDG 13.b** Climate planning for vulnerable communities.	***Stakeholder Engagement (custom):*** Measures inclusive participation in planning and local adaptation processes.	**GRI 413-1 (Operations with local community engagement):** Demonstrates meaningful community participation; climate justice and CBA principles.
Local Languages and Climate Topics Supported: Number of languages, dialects, and climate-risk topics available through VosynCore, including Sinhala, Tamil, English, flood preparedness, heat risk, landslide awareness, and drought guidance.	**SDG 13.3** Climate education and awareness. **SDG 13.b** Capacity-building for vulnerable communities.	**Product Quality/Accessibility:** Measures accessibility of the solution for underserved language groups.	**GRI 417 (Product/service information):** Ensures information is provided in languages users understand; accessibility and inclusion reporting.
Facilitators and Response Partners Trained: Number of volunteers, educators, women-led group members, health workers, local officers, or disaster-response partners trained to use VosynCore for climate	**SDG 13.1** Resilience and adaptive capacity. **SDG 13.3** Capacity-building and awareness.	**Employees/Stakeholders Trained (adapted IRIS metric):** Counts capacity-building among community and partner facilitators.	**GRI 404-1 (Training and education):** Reports training provided; emergency preparedness and capacity-building indicators.

communication.

Table 1: VosynCore SDG 13 Performance Matrix, including KPI mapping, UN indicator harmonization, and climate-action impact pathways.

By establishing this standardized, transparent KPI framework, VosynCore bridges the operational gap between multilingual climate communication and measurable preparedness outcomes.

For households, women-led groups, schools, health institutions, community organizations, local officers, disaster volunteers, and impact investors, this matrix turns climate communication into a measurable and auditable pathway for resilience.

For more information on the compliance frameworks that govern our operations, visit our ***Compliance Document.***

Impact Pathway

VosynCore's Theory of Change (TOC) for SDG use cases outlines how inputs and activities translate into outputs, outcomes, and long-term impact. For SDG 13, the pathway begins with accessible multilingual climate information and leads toward stronger community preparedness, adaptation literacy, and resilience.

Learning Agenda, Pilot Roadmap & Scalability Strategy

Sri Lanka Climate Risk Communication and Preparedness Pilot

VosynCore's SDG 13 initiative is not a one-off deployment; it is conceived as a learning journey that tests how multilingual voice guidance can improve climate-risk comprehension, preparedness behaviour, and community adaptation participation.

Learning Agenda

We have formulated key learning questions to guide our SDG 13 pilot and inform future scaling. These questions focus on warning comprehension, climate literacy, preparedness actions, community adaptation, gender-responsive communication, and trust in verified information.

1. Effectiveness in Climate Warning Comprehension: How much does multilingual voice guidance improve users' understanding of flood, landslide, heat, drought, and rainfall warnings?

- **Goal:** Compare comprehension, recall, and confidence between users receiving standard climate-risk messages and users receiving VosynCore-supported explanations.
- **Example:** Is there a measurable increase in users who correctly identify what a flood warning means and what household actions should be taken before the event?

2. Improvement in Preparedness Behaviour: Does VosynCore help households complete practical preparedness tasks before climate-related hazards occur?

- **Goal:** Measure changes in self-reported and observed preparedness actions, including document protection, medication checks, emergency bags, safe-route awareness, and family communication plans.

- **Example:** Do users become more confident preparing older relatives, children, school materials, and household supplies after receiving localized voice guidance?

3. Trust, Inclusion, and Community Adaptation Participation: Can VosynCore make adaptation planning more inclusive by helping women, youth, Tamil-speaking users, and low-literacy users understand and contribute to climate planning?

- **Goal:** Evaluate whether translated meeting materials, spoken summaries, and two-way question support improve participation in community adaptation activities.

- **Example:** Do women-led groups and local volunteers report greater confidence sharing preparedness information and collecting community feedback after using VosynCore?

To answer these questions, our Monitoring, Evaluation, and Learning (MEL) team will use a mixed-methods approach: pre/post user surveys, comprehension checks, community interviews, simulation exercises, facilitator feedback, and task-completion tracking.

For more information, please visit our ***"Impact Pathway" document.***

Pilot Roadmap

The pilot will proceed in iterative stages to ensure that VosynCore is safely and responsibly adapted to local climate-risk, disaster-preparedness, and community-adaptation contexts.

Months 0-3 (Preparation and Co-Design): Establish Partnerships and Foundations

- Finalize agreements with community organizations, women-led groups, schools, local officers, disaster-response partners, health institutions, and climate-adaptation stakeholders.

- Conduct co-design workshops with households, women and girls, youth groups, older residents, Tamil-speaking users, Sinhala-speaking users, local volunteers, and community leaders.

- Prepare localized Sinhala, Tamil, and English content libraries for flood preparedness, landslide awareness, heat-risk guidance, drought preparedness, emergency hygiene, climate literacy, and adaptation planning.

- **Milestones:** Secure partner approvals; configure the initial VosynCore SDG 13 knowledge base; complete safety review for climate and disaster guidance; design baseline survey and simulation tools.

Months 4-9 (Implementation and Monitoring): Field Deployment and Feedback Loops

- Roll out VosynCore-supported access points through community centres, schools, local organizations, women-led groups, disaster-preparedness sessions, mobile outreach, and partner-led awareness activities.
- Implement real-time monitoring of translation accuracy, user comprehension, requested topics, completion of preparedness tasks, and common points of confusion.
- Facilitate monthly learning circles with households, women and girls, community volunteers, local officers, health workers, educators, and adaptation stakeholders to refine content and improve trust.
- **Milestones:** Achieve target engagement levels within the pilot cohort; collect evidence on warning comprehension, preparedness behaviour, and climate-literacy gains; improve content based on feedback.

Months 10-12 (Evaluation and Scaling Prep): Impact Assessment and Synthesis

- Execute endline surveys, usage analysis, household interviews, facilitator feedback sessions, simulation reviews, and partner assessments to evaluate the pilot's effectiveness.
- Synthesize findings into a Climate Risk Communication and Preparedness Playbook for Sri Lanka, including tested voice scripts, source-verification rules, escalation pathways, and partner workflow recommendations.
- Refine AI models, content libraries, OCR workflows, and escalation rules based on evaluation results, especially for urgent warnings, health-sensitive guidance, and vulnerable-group communication.
- **Milestones:** Publish a validated SDG 13 impact report; finalize the roadmap for Phase 2 scaling; identify institutional partners for broader adoption.

Scalability Strategy

Looking beyond the pilot, our ambition is to scale vertically by embedding VosynCore into climate-risk communication and horizontally by adapting the model across hazards, regions, languages, and community institutions.

- **Community and School Integration:** We aim to embed VosynCore into environmental clubs, community centres, school awareness programmes, women-led groups, and local climate preparedness campaigns.
    - **Example:** A school or community group could use VosynCore to deliver Sinhala, Tamil, and English voice guidance on flood preparedness, heat safety, climate literacy, and household emergency planning.

- **Hazard Replicability:** The modular architecture allows new climate-risk domains to be added over time, including floods, landslides, heat stress, drought, coastal flooding, and post-disaster health precautions.
    - **Example:** Once the model is stabilized for flood preparedness and climate literacy, the same interaction design can be adapted for landslide warnings, drought preparedness, and heat-risk communication.
- **API-First Partnership Model:** To achieve exponential growth, we will offer VosynCore's translation and explanation capabilities to NGOs, disaster-response groups, education partners, health institutions, and local government channels.
- **Example:** A climate-alert system, school climate module, community WhatsApp bot, or disaster-preparedness registration form could call VosynCore to deliver the same trusted guidance in multiple languages.
- **Gender-Responsive Adaptation Support:** Evidence from Gampaha shows that women and girls carry major household and community responsibilities during floods, yet may be insufficiently included in formal planning.
    - **Example:** Women-led groups, health volunteers, and youth facilitators could be trained to use VosynCore for preparedness discussions, household checklists, and feedback collection in Sinhala and Tamil.

This strategy ensures that VosynCore evolves from a localized climate communication tool into a replicable platform for adaptation literacy and preparedness across Sri Lanka and similar multilingual contexts.

Modelled Projections

As VosynCore moves toward field deployment, we have developed a series of evidence-informed projections based on the barriers identified in Sri Lankan climate literacy, community adaptation, flood experience, and disaster communication studies, including awareness-action gaps, grassroots participation barriers, and fragmented emergency communication.

Projected Outcomes (Short to Medium Term):

In the initial 12-24 months of deployment, our focus is on establishing a baseline evidence trail for multilingual climate-risk communication and preparedness guidance.

- **Improved Climate Warning Understanding:** A 35%-45% projected increase in users correctly interpreting flood, landslide, heat, drought, and rainfall warnings after using VosynCore.
- **Higher Preparedness Task Completion:** Up to 30% projected improvement in users' ability to identify household preparedness steps, protect important items, prepare emergency supplies, and understand safe-response actions.

- **Improved Climate Literacy Confidence:** A measurable increase in residents, students, and community facilitators who can explain climate risks, adaptation options, and preparedness steps in everyday language.

- **Better Emergency Communication Coordination:** Reduced confusion during simulations through clearer translated guidance, standardized messages, and improved communication between community actors and response institutions.

Projected Outcomes (Medium to Long Term):

Looking toward 2030, the integration of VosynCore into climate-risk communication could support broader changes in adaptation literacy, community preparedness, and resilience planning.

- **Stronger Community Preparedness:** Households become more willing and able to act before hazards escalate because warning messages are easier to understand and linked to practical actions.

- **Expanded Gender-Responsive Climate Action:** Women, girls, and caregivers receive clearer preparedness information and stronger pathways to participate in local adaptation planning.

- **Improved Trust in Verified Climate Information:** Users rely more on trusted warnings and partner-validated guidance rather than scattered rumours, unclear social media posts, or delayed informal explanations.

- **Scalable Model for Climate Adaptation Communication:** VosynCore can become a reusable approach for multilingual early warning, climate education, preparedness, and adaptation communication in Sri Lanka and other multilingual developing contexts.

Conclusion: High-Potential Innovation for SDG 13

VosynCore fuses innovative AI with practical climate communication to address one of the most persistent barriers in climate action: the gap between knowing that climate risks exist and knowing what to do before, during, and after a hazard.

This initiative demonstrates how inclusive AI can amplify the impact of existing warning systems, climate education programmes, adaptation plans, health preparedness activities, and community organizations by making their guidance understandable, actionable, and trusted in local languages.

Key Takeaways:

- **Systemic Force Multiplier:** VosynCore does not replace climate agencies, disaster authorities, health institutions, or local leaders; it unlocks their value by helping users understand and act on verified guidance.

- **Measurable Social Return on Investment (SROI):** The platform converts climate communication into trackable indicators such as warning comprehension, preparedness-task completion, climate literacy confidence, and participation in adaptation planning.

- **Distinct SDG 13 Focus:** This use case is intentionally framed around climate risk communication, early warning comprehension, preparedness, adaptation literacy, and community climate action, keeping it separate from SDG 1 poverty relief, SDG 7 clean energy, SDG 11 municipal services, and SDG 6 water safety.

- **Rigorous Institutional Alignment:** Built for accountability, the platform's roadmap connects SDG 13 targets with KPI tracking, community learning, safety safeguards, and evidence-based scaling.

Achieving SDG 13 requires more than climate awareness. People need clear, trusted, and practical ways to understand warnings, prepare households, participate in adaptation, and respond safely to climate-related hazards.

The SDG 13 initiative offers a high-potential innovation that demonstrates the power of multilingual AI to turn climate information into everyday preparedness and community resilience.

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Additional climate-action literature on early warning systems, heat stress, drought preparedness, coastal adaptation, landslide risk, and local climate governance should be incorporated during pilot validation.

Pilot Evidence Validation Notes

The SDG 13 pilot should treat all projections as working assumptions until they are tested with real users, women-led groups, community organizations, schools, health institutions, and disaster-response partners.

Climate and disaster guidance should include strict safeguards. VosynCore should prioritize verified warnings and avoid unsupported emergency, medical, engineering, or evacuation advice beyond approved partner content.

Flood, heat, landslide, and drought communication should be tested with diverse households, women and girls, older residents, students, Tamil-speaking users, Sinhala-speaking users, and low-literacy users.

Community adaptation guidance should be validated with trusted partners so that users receive realistic options and understand when to contact local authorities, health workers, or emergency services.

Climate-action monitoring should assess whether multilingual guidance improves trust in verified information, reduces confusion, supports preparedness behaviour, and strengthens long-term community adaptation rather than only short-term awareness.